



FITNESS TRACKER

Database Management
System Project

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PROBLEM STATEMENT

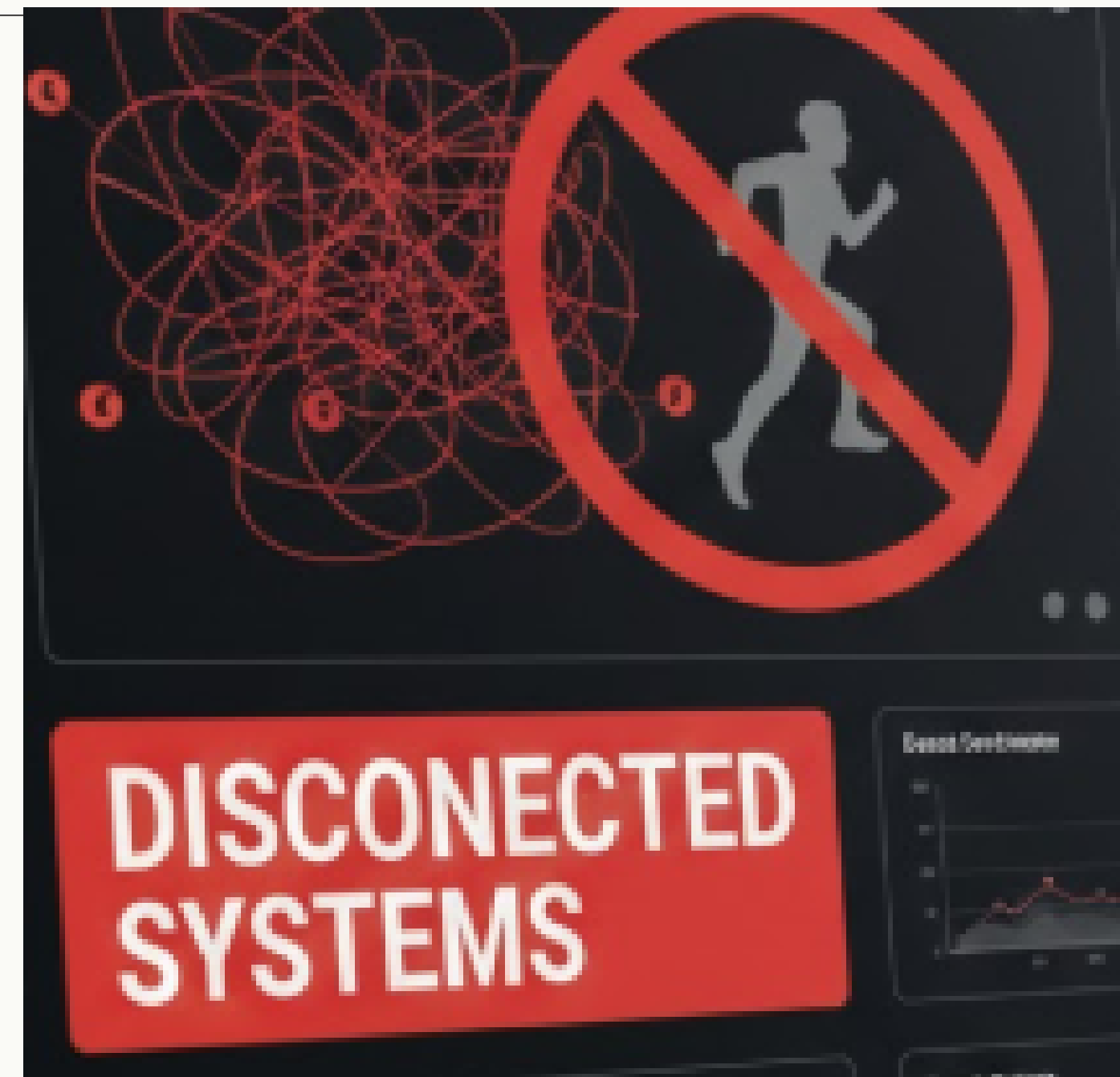
Problem

Modern fitness applications collect large volumes of user health data such as activity tracking, workouts, heart rate, sleep patterns, and nutrition records

However, this information is often stored in disconnected systems, making it difficult to maintain structured relationships between data, analyze user performance effectively, and provide timely health insights or automated responses.

OBJECTIVE

THE OBJECTIVE OF THIS PROJECT IS TO DESIGN A CENTRALIZED FITNESS TRACKER DATABASE THAT ORGANIZES FITNESS DATA USING A RELATIONAL STRUCTURE, ENABLING EFFICIENT STORAGE, MEANINGFUL DATA RELATIONSHIPS, AND AUTOMATED MONITORING FOR IMPROVED HEALTH TRACKING AND ANALYSIS.



KEY FEATURES

USER
MANAGEMENT

DEVICE
INTEGRATION

WORKOUT
TRACKING

HEALTH
MONITORING

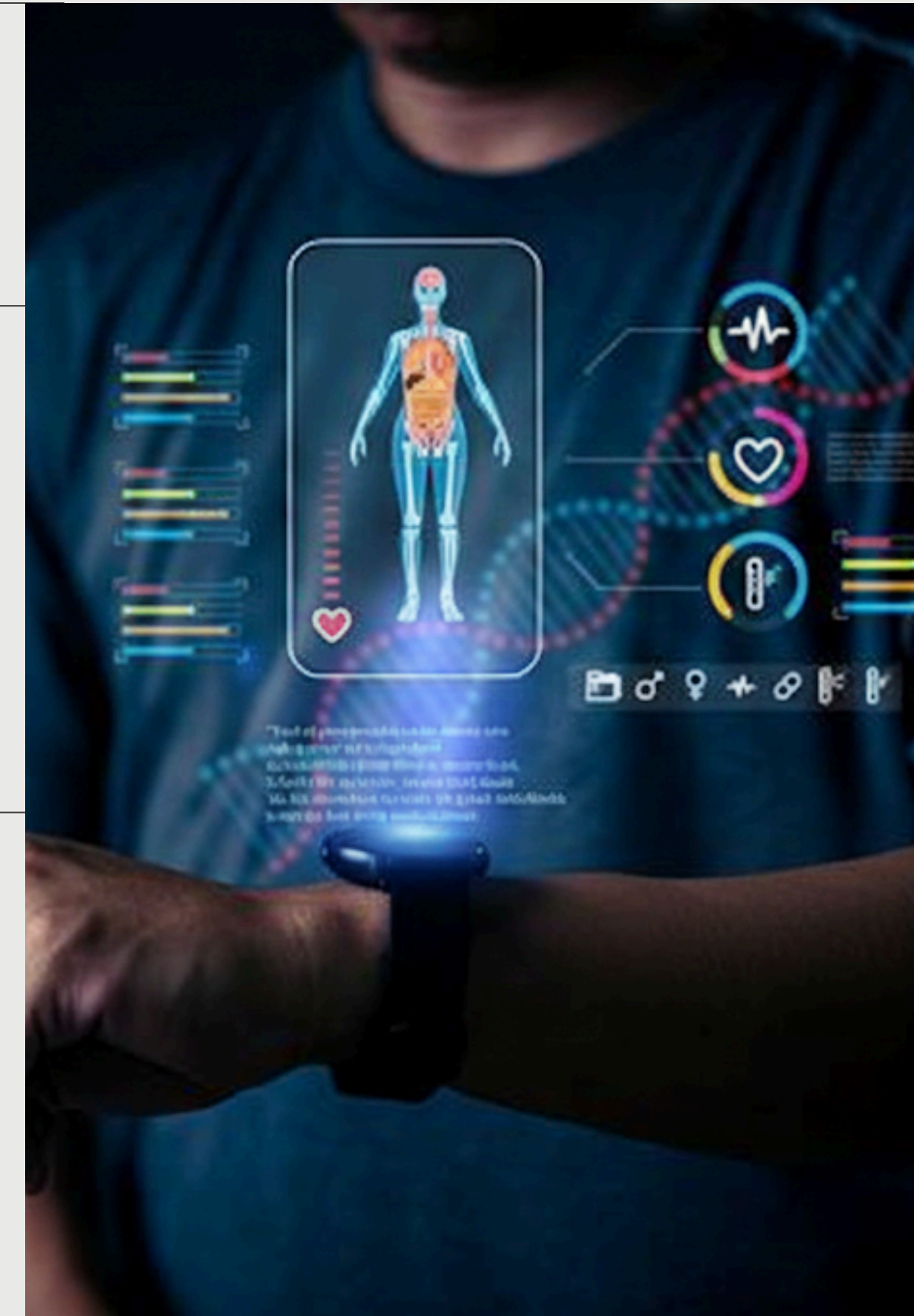
SMART
AUTOMATION

EMERGENCY
ALERTS

ACHIEVEMENTS
SYSTEM

DATA
ANALYTICS

The Fitness Tracker Database provides a structured platform to manage user fitness activities, wearable device data, workouts, and health records in a centralized system. It enables automated badge rewards, emergency health alerts, and personalized recommendations while supporting analytical insights for monitoring performance and improving overall fitness management.



TECHNOLOGY USED

DATABASE

- MYSQL

TOOLS

- MYSQL WORKBENCH
- CHAT GPT
- CANVA(PRESENTSTION)

CONCEPTS IMPLEMENTED

- Relational Database Design
- Normalization (3NF)
- Primary & Foreign Keys
- Constraints

ADVANCE SQL

- JOIN Operations
- Aggregate Functions
- Triggers
- Stored Procedures

ENTITIES

CORE ENTITIES

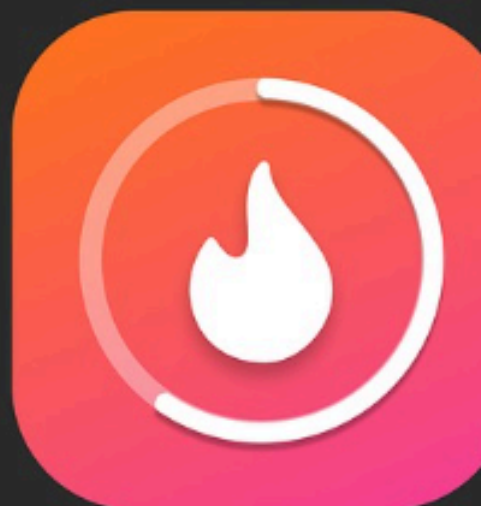
- Users
- Devices
- Exercises
- Subscriptions
- Features
- Badges
- Challenges

RELATIONSHIP ENTITIES

- User Devices
- Workout Logs
- Workout Exercises
- User Subscriptions
- User Badges
- User Challenges

LOG ENTITIES

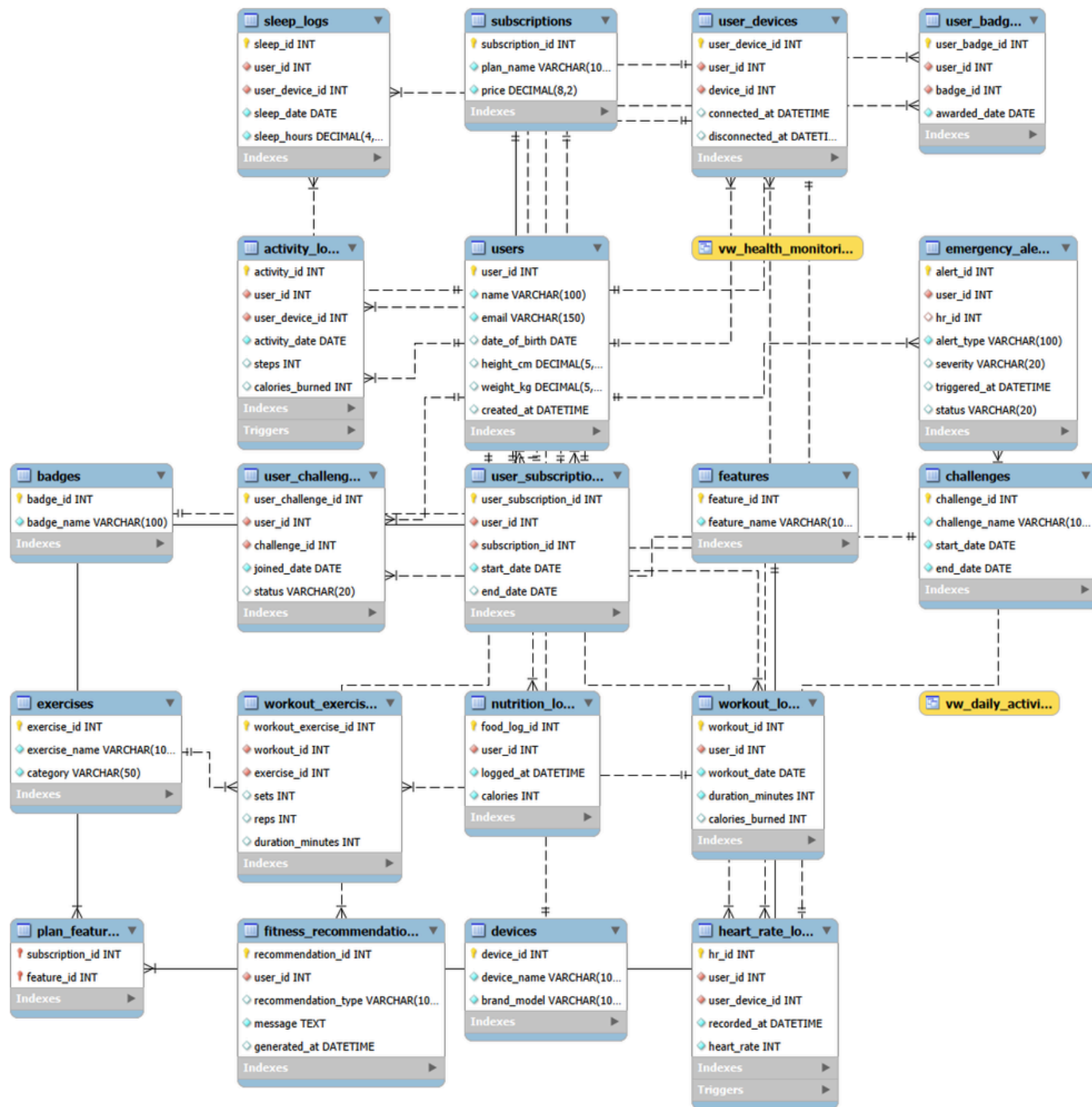
- Activity Logs
- Heart Rate Logs
- Sleep Logs
- Nutrition Logs
- Emergency Alerts
- Fitness Recommendations



ER DIAGRAM

THE ER DIAGRAM REPRESENTS RELATIONSHIPS BETWEEN ENTITIES:

- ONE USER → MANY WORKOUTS
- ONE USER → MANY DEVICES
- ONE SUBSCRIPTION → MANY FEATURES
- WORKOUTS → MULTIPLE EXERCISES
- HEALTH LOGS GENERATED VIA DEVICES



SYSTEM WORKING DEMO

DATABASE CREATED

Navigator

HEMAS

Filter objects

accommodation

fitness_tracker

Tables

Views

Stored Procedures

Functions

retail

sys

TABLES GRENERATED

Tables_in_fitness_tracker

activity_logs

badges

challenges

devices

emergency_alerts

exercises

features

fitness_recommendations

heart_rate_logs

nutrition_logs

plan_features

sleep_logs

subscriptions

user_badges

user_challenges

user_devices

user_subscriptions

QUERY RUN

-- 2.USER WITH ACTIVE SUBSCRIPTIONS

SELECT u.name, s.plan_name

FROM users u

JOIN user_subscriptions us ON u.user_id = us.user_id

JOIN subscriptions s ON us.subscription_id = s.subscription_id

WHERE us.end_date >= CURDATE()

-- 3.USERS WITHOUT SUBSCRIPTIONS

SELECT u.name

FROM users u

LEFT JOIN user_subscriptions us ON u.user_id = us.user_id

WHERE us.user_id IS NULL

Result Grid

Filter Rows:

Export:

name	plan_name
Aarav Sharma	Premium
Anil Shrivastava	Premium

TRIGGER

INITIAL BADGE COUNT

Result Grid

Filter Rows:

Edit:

user_badge_id	user_id	badge_id	awarded_date
1	1	1	2026-01-15
2	3	2	2026-01-16
NULL	NULL	NULL	NULL

USER COMPLETES 10000+ STEPS

INSERT INTO activity_logs

(user_id, user_device_id, activity_date, steps, calories_burned)

VALUES

(2, 2, CURDATE(), 12000, 450);

BADGE COUNT INCREASES

Result Grid

Filter Rows:

Edit:

user_badge_id	user_id	badge_id	awarded_date
1	1	1	2026-01-15
2	3	2	2026-01-16
3	2	1	2026-02-23
NULL	NULL	NULL	NULL

8

PRESENTED BY: ABHASHREE SISODIA

FUTURE ENHANCEMENTS



TECHNICAL IMPROVEMENTS

- Mobile App Integration
- Real-time API connectivity
- Cloud database deployment

AI ENHANCEMENTS

- Personalized workout prediction
- Diet recommendation engine
- Health risk prediction

FUTURE EXPANSION

- Social fitness groups
- Trainer dashboard
- GPS activity tracking

CONCLUSION

PROJECT SUMMARY

THIS PROJECT SUCCESSFULLY:

- DESIGNED A NORMALIZED FITNESS DATABASE
- MODELED REAL-WORLD FITNESS TRACKING SYSTEMS
- IMPLEMENTED AUTOMATION USING TRIGGERS
- GENERATED ANALYTICAL HEALTH REPORTS

KEY LEARNING OUTCOMES

- DATABASE DESIGN PRINCIPLES
- RELATIONSHIP MODELING
- SQL OPTIMIZATION
- AUTOMATION USING PROCEDURES



THANK YOU

